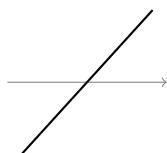


Zeros, Multiplicity, and Graph Behavior

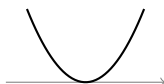
Algebra 2 Polynomial Factoring, Roots, and Graphs

Factoring to find zeros, reading multiplicity as crossing or touching, describing end behavior, and building a polynomial from its zeros

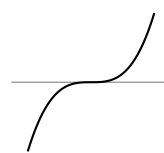
What multiplicity looks like on a graph. A factor's exponent tells you the shape of the curve where it meets the x -axis: an odd exponent passes through, an even exponent bounces off, and a higher odd exponent flattens as it passes.



odd exponent: crosses



even exponent: touches



higher odd: flattens

The three local shapes at a zero. No values shown: use the factors given in each problem.

Directions.

- Factor completely before reading anything off a polynomial — the factored form is what shows the zeros and their multiplicities.
- When a problem asks for behavior, name the zero, its multiplicity, and whether the curve crosses or touches.
- End behavior is set by the degree (odd or even) and the sign of the leading coefficient.

1. Factor $f(x) = x^3 - 4x$ completely, then list every real zero of f . Write the factored form on the answer line.

Answer: _____

- 2.** A polynomial is given in factored form:

$$g(x) = (x + 1)(x - 3)^2$$

List each real zero with its multiplicity, and state whether the graph of g crosses or touches the x -axis at each one. Write the zeros on the answer line.

Answer: _____

- 3.** Factor $h(x) = x^3 - 3x^2 - 10x$ completely, then list every real zero of h . Write the factored form on the answer line.

Answer: _____

4. Describe the end behavior of $f(x) = -2x^3 + 5x - 1$ by completing both statements, and say in one sentence which two features of the polynomial decided them.

as $x \rightarrow \infty$, $f(x) \rightarrow$ _____ as $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Answer: _____

5. A polynomial is given in factored form:

$$p(x) = x^2(x - 4)^3$$

State the degree of p , list each real zero with its multiplicity, and say whether the graph crosses, touches, or flattens as it passes at each zero. Write the zeros on the answer line.

Answer: _____

6. An engineer needs a curve that meets the x -axis exactly at $x = -2$ and at $x = 1$, passing straight through at $x = -2$ but bouncing off at $x = 1$. Write the polynomial of least degree with leading coefficient 1 that does this, in expanded form.

Answer: _____

7. Solve $x^4 - 5x^2 + 4 = 0$ by factoring. (Hint: the left side is a quadratic in x^2 .) List all four real zeros on the answer line.

Answer: _____

8. Describe the end behavior of $g(x) = 3x^4 - x^3$ by completing both statements, and say in one sentence why both ends behave the same way here but not in problem 4.

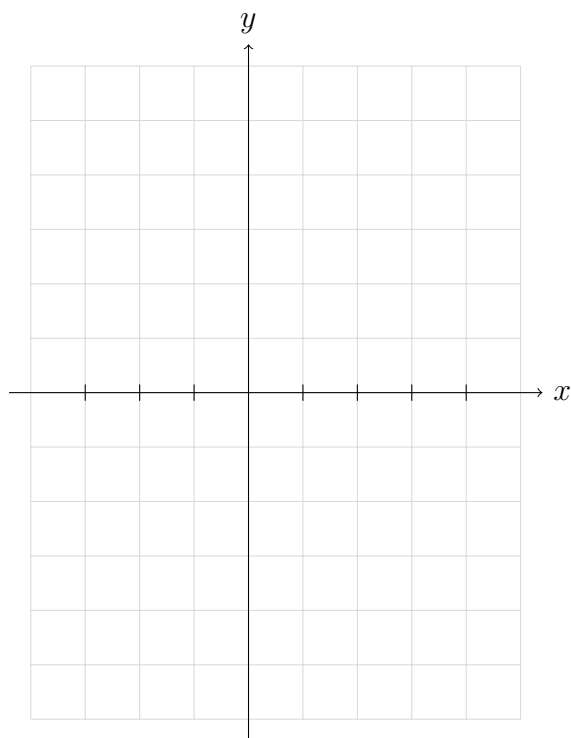
as $x \rightarrow \infty$, $g(x) \rightarrow$ _____ as $x \rightarrow -\infty$, $g(x) \rightarrow$ _____

Answer: _____

9. A designer sketches a profile curve that crosses the x -axis at $x = -3$ and just touches it at $x = 2$, with no other x -intercepts. Write the polynomial of least degree with leading coefficient 1 that matches the sketch, in expanded form.

Answer: _____

10. Sketch the graph of $y = x(x + 2)^2(x - 3)$ on the grid below. Mark every x -intercept, label each one as crossing or touching, and make both ends of your sketch match the end behavior. Each square is one unit wide; choose your own vertical scale and label it. Write the zeros on the answer line.



Answer: _____